Physics 4700H: Assignment VI

Jeremy Favro (0805980)
Trent University, Peterborough, ON, Canada

October 20, 2026

Problem 1. Consider a system which can be in one of three states, corresponding to energies ξ_1, ξ_2, and ξ_3. What is the partition function of this system?

Solution 1.

$$Z = \sum_i p_i = \sum_i e^{\xi_i/k_BT} = e^{\xi_1/k_BT} + e^{\xi_2/k_BT} + e^{\xi_3/k_BT}.$$

Problem 2. In class we found that for a two state system $F = -k_BT\ln\left(1+e^{-\epsilon/k_BT}\right)$. What is the entropy of this system as a function of temperature? Consider the value of your result at $T = 0$ and $T \to \infty$. Comment on whether the values you get make sense.

Solution 2. Recall that $dF = dU - TdS - SdT$. In the microcanonical ensemble (reservoir plus system),

$$-\frac{\partial F}{\partial T}\Big)_{N,V,U,S} = S.$$

Then we just differentiate the expression we have for F,

$$S = k_B\ln\left(1+e^{-\epsilon/k_BT}\right) + \frac{k_BT}{1+e^{-\epsilon/k_BT}}\left[\frac{\epsilon}{k_BT^2}e^{-\epsilon/k_BT}\right] = k_B\ln\left(1+e^{-\epsilon/k_BT}\right) + \frac{\epsilon e^{-\epsilon/k_BT}}{\left[1+e^{-\epsilon/k_BT}\right]T}.$$

In the $T = 0$ case, $e^{-\epsilon/k_BT} \to e^{-\infty} \to 0$, so the first term goes to zero as $\ln(1) = 0$. The second term's numerator goes to zero, as does the denominator, but the denominator is bounded by

$$T \leq \left[1+e^{-\epsilon/k_BT}\right]T,$$

so the entire fraction is bounded by

$$\frac{\epsilon e^{-\epsilon/k_BT}}{\left[1+e^{-\epsilon/k_BT}\right]T} \leq \frac{\epsilon e^{-\epsilon/k_BT}}{T}$$

(reciprocal flips the inequality, if you want) which certainly goes to zero as exponentials beat polynomials. Then since the original expression is less than something going to zero but is itself greater than zero, $S(0) \to 0$.

In the $S(T \to \infty)$ case, $e^{-\epsilon/k_BT} \to e^0 \to 1$ so the first term goes to $k_B\ln(2)$ and the second goes like $1/T$, which goes to zero at large T, hence $S(T \to \infty) = k_B\ln(2)$. The first value certainly makes sense, it is a law of thermodynamics that $S(0) \to 0$ in most systems. The second result is consistent with the Gibbs entropy provided the two states are equally likely,

$$S_G = -k_B\sum_i p_i\ln p_i = k_B\left[\frac{\ln(2)}{2} + \frac{\ln(2)}{2}\right]$$

where I pulled the negative into the ln to flip the probability, and took $p_i = 1/2$. I'm not totally sure this is the correct understanding of what's happening but it works.

Problem 3. Replacing sums by integrals

(a) Show that $\sum_{n=0}^{\infty} u^n = 1/(1-u)$ for $u < 1$.

(b) Consider the function $f(n) = e^{-Cn}$ where n is a non-negative integer. Make a sketch showing $f(n)$ for $C = 0.1$, 1 and 10.

(c) Calculate and compare

$$S_{10} = \sum_{n=1}^{\infty} e^{-10n} \quad \text{and} \quad I_{10} = \int_0^{\infty} e^{-10x} dx.$$

(d) Calculate and compare

$$S_{0.1} = \sum_{n=1}^{\infty} e^{-0.1n} \quad \text{and} \quad I_{0.1} = \int_0^{\infty} e^{-0.1x} dx.$$

(e) What does this teach us about replacing sums with integrals?

Solution 3.

(a) Consider

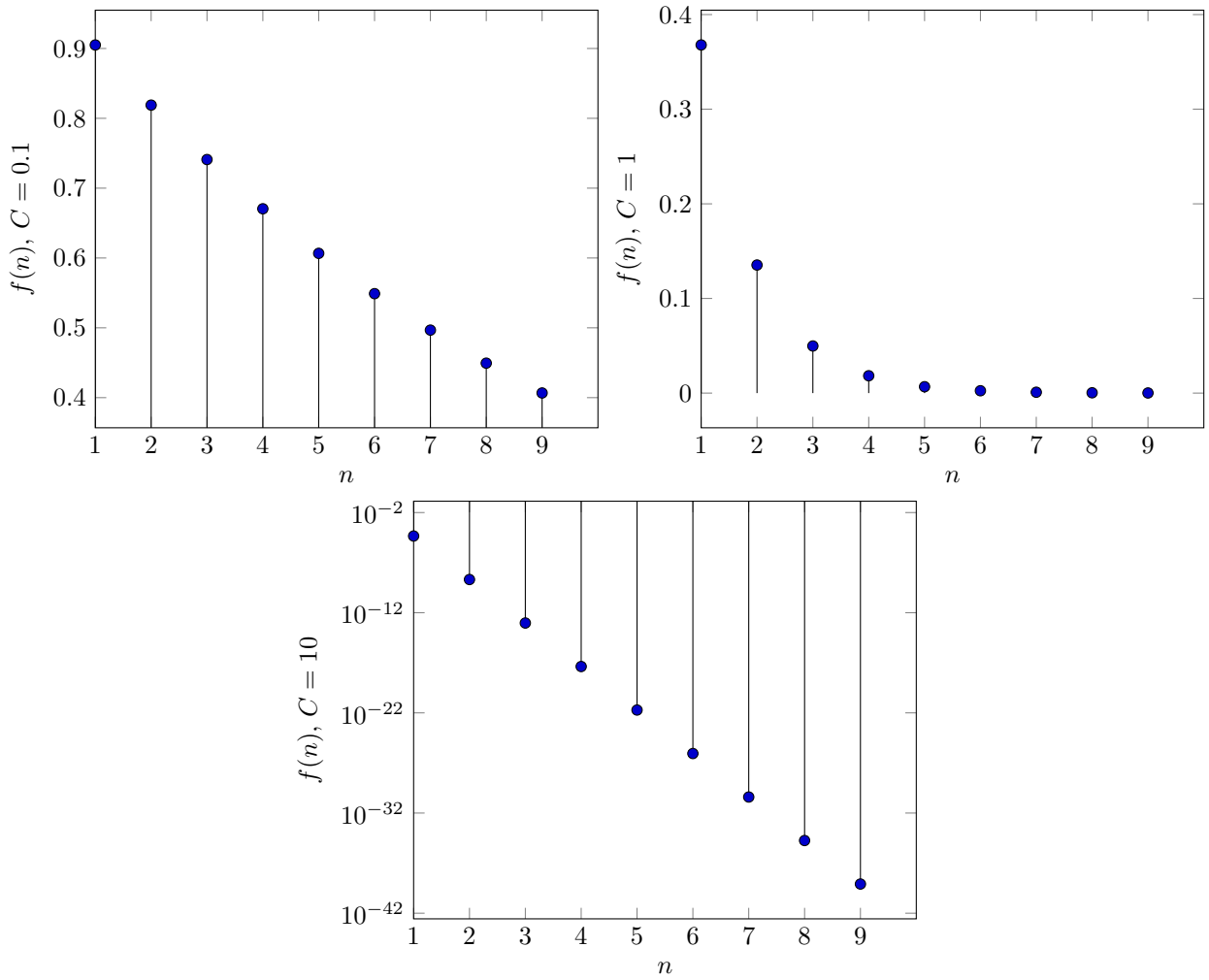
$$S = \sum_{n=0}^{\infty} u^n = 1 + u + u^2 + u^3 + \dots$$

and

$$Su = u + u^2 + u^3 + u^4 + \dots.$$

Then $S - Su = 1 \implies S = 1/(1-u)$.

(b) This isn't really a sketch, but it serves the purpose.



(c) Using the geometric series identity we proved in (a),

$$S_{10} = \sum_{n=1}^{\infty} e^{-10n} = \sum_{n=1}^{\infty} (e^{-10})^n = \frac{1}{1 - e^{-10}} \approx 1.00004540199,$$

which is perfectly fine since $e^{-10} \ll 1$. Then

$$I_{10} = \int_0^{\infty} e^{-10x} dx = -\frac{1}{10} \left[e^{-\infty} - e^0 \right] = \frac{1}{10}$$

(d) Again using the geometric series identity since $e^{-0.1} < 1$, $S_{0.1} = 1/(1 - e^{-0.1}) \approx 10.5$ and

$$I_{0.1} = -\frac{1}{0.1} \left[e^{-\infty} - e^0 \right] = 10.$$

(e) This displays what we saw in class, the step size (in both cases above, $\Delta x = 1$) is too big compared to the length scales over which e^{-10n} is changing providing a poor approximation, but is fine over the scale of $e^{-0.1n}$ and provides a good approximation.